

Inumula Teja

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[GitHub](#) | [LeetCode](#) | [Codeforces](#) | [LinkedIn](#)

Education

Sardar Vallabhbhai National Institute of Technology, Surat

Bachelor of Technology in Artificial Intelligence

Aug 2024 – May 2028 (Expected)

CGPA: 7.72 (Till Sem 4)

Projects

MediCare – Hospital Management System | [GitHub](#)

Apr 2024

Technologies: Python, Django, SQLite, HTML5, CSS3, JavaScript

- Developed the full-stack architecture of a healthcare portal using Django and SQLite, designing a relational database schema to manage departments, staff roles, and patient records.
- Implemented a scheduling conflict resolution algorithm in Python utilizing timezone-aware objects to query and flag overlapping appointments within a ± 59 -minute window.
- Optimized database query performance by utilizing Django's `select_related()` to eagerly load doctor and staff profiles, mitigating the N+1 query problem.
- Engineered secure authentication and validation workflows, customizing Django's `UserCreationForm` to enforce strict server-side sanitization, email uniqueness constraints, and password complexity criteria.
- Programmed a consolidated search engine using Django Q objects to perform complex textual queries across multiple database tables (Staff, Specialities, Therapies), dynamically decorating and sorting results.

Topology-Aware Sepsis Prediction (CAT3P) | [GitHub](#)

June 2026

Technologies: Python, PyTorch, DuckDB, MIMIC-IV

- Developed a deep learning framework for four-stage sepsis severity prediction (No Sepsis, Sepsis, Severe Sepsis, Septic Shock) using temporal electronic health records from the MIMIC-IV dataset.
- Designed a topology-aware Graph Neural Network (CAT3P) to model interactions among organ systems while capturing temporal disease progression.
- Built an end-to-end preprocessing pipeline featuring Sepsis-3 label generation, hourly time-series alignment, missing-value imputation, feature normalization, and graph tensor construction.
- Improved minority-class prediction using class-weighted cross-entropy loss and weighted random sampling, achieving a 57.8% Macro F1-score and 0.8869 Macro AUROC.

Technical Skills

Languages: C++ , C, Python, MySQL, HTML, CSS, JavaScript

Web Technologies: React JS, Node.js, Django

AI/ML: NumPy, Pandas, Machine Learning

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Architecture, Computer Networks,

Tools & Platforms: Linux (Ubuntu), Git, GitHub, VS Code, Jupyter Notebook

Achievements

- Winner — Squidathon (Cultural Fest) | Secured 1st place among competing teams in the flagship strategic event organized by the Chemical Engineering Society (ChES), SVNIT, demonstrating collaborative problem-solving under pressure.
- LeetCode | Solved 245+ algorithmic problems.
- Codeforces | achieved a peak rating of 1205 (Newbie rank) by optimizing data structures and algorithms under time constraints.
- Inter-NIT Kho Kho | Participated in the Inter-NIT Kho Kho Tournament, representing SVNIT Surat, demonstrating teamwork, discipline, and competitive spirit.